

PRACTICAL 9

Aim

To perform social media trend detection using Apache Spark with PySpark by analyzing hashtag frequency, engagement, categories, regions, and daily trends.

The practical focuses on creating a social media hashtag dataset, loading the data into a Spark DataFrame, examining its schema and records, calculating hashtag frequency, identifying top trending hashtags, analyzing trends by category and region, comparing trends across multiple days, generating a trend analysis report, and visualizing hashtag trends.

Problem Definition

Social media platforms such as Twitter/X, Instagram, Facebook, and LinkedIn generate a massive amount of data every day. Users continuously post content related to current events, technology, sports, entertainment, politics, business, and other topics.

A social media analytics company needs to identify popular and emerging topics from this large volume of data. Traditional data processing tools may become inefficient when processing large-scale social media datasets.

Apache Spark provides distributed DataFrame operations and aggregation functions that can efficiently process large datasets. Spark can be used to analyze hashtag frequency, engagement, categories, regions, and trends over time.

The task is to create or obtain a social media hashtag dataset and perform trend detection using PySpark.

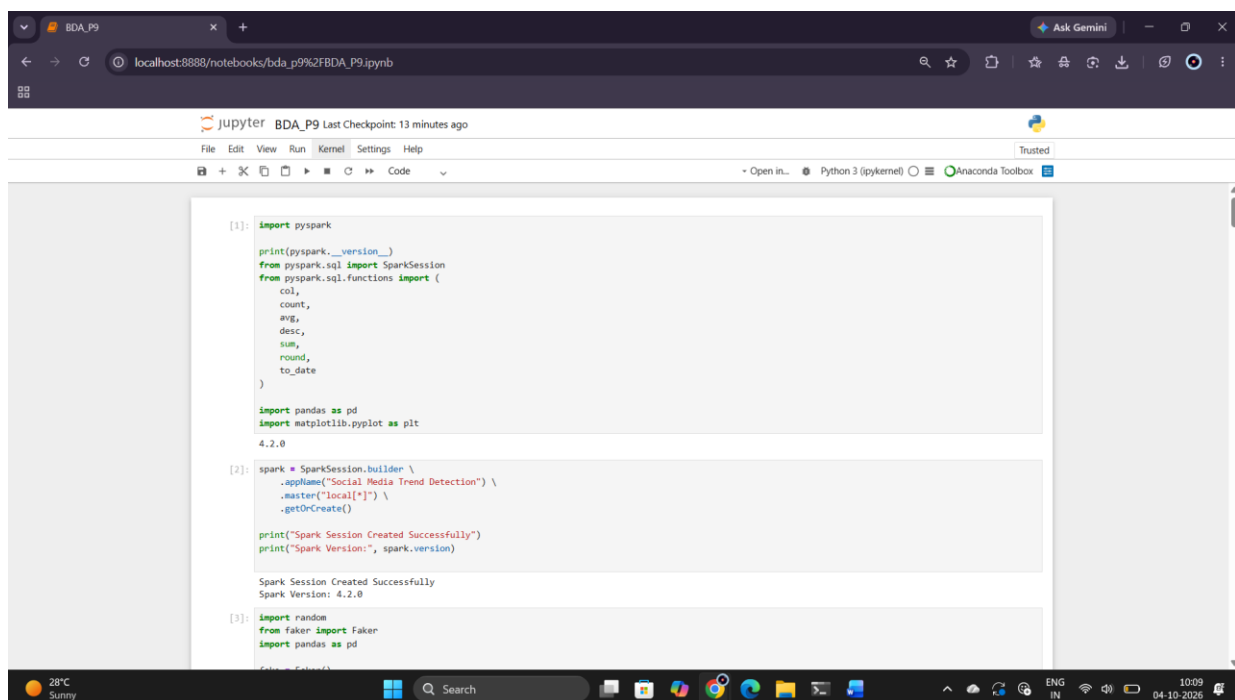
The practical includes creating a SparkSession, loading and displaying the social media dataset, examining the schema, validating the dataset, counting hashtag frequency, identifying top trending hashtags, analyzing hashtag popularity by category and region, comparing trends across multiple days, generating a trend analysis report, and visualizing the results.

Objectives

- Understand social media analytics using Apache Spark.
- Create and configure a SparkSession.
- Create or obtain a social media hashtag dataset.
- Load social media records into a Spark DataFrame.
- Display and examine the social media dataset.
- Display the schema of the DataFrame.
- Validate the dataset for missing values and incorrect records.

- Calculate hashtag frequency.
- Identify the most popular hashtags.
- Identify the top 3 trending hashtags.
- Analyze hashtag trends by category.
- Compare Technology, Sports, and Entertainment trends.
- Analyze hashtag popularity by region.
- Compare hashtag trends across multiple days.
- Analyze hashtag engagement.
- Calculate a trend score using frequency and engagement.
- Generate a social media trend analysis report.
- Visualize hashtag trends using charts.
- Understand the importance of Spark DataFrame operations in social media analytics.
- Analyze emerging social media trends using PySpark.

Output



The screenshot displays a Jupyter Notebook environment with the following code cells:

```
[1]: import pyspark

print(pyspark.__version__)
from pyspark.sql import SparkSession
from pyspark.sql.functions import (
    col,
    count,
    avg,
    desc,
    sum,
    round,
    to_date
)

import pandas as pd
import matplotlib.pyplot as plt
4.2.0
```

```
[2]: spark = SparkSession.builder \
    .appName("Social Media Trend Detection") \
    .master("local[*]") \
    .getOrCreate()

print("Spark Session Created Successfully")
print("Spark Version:", spark.version)

Spark Session Created Successfully
Spark Version: 4.2.0
```

```
[3]: import random
from faker import Faker
import pandas as pd
```

The notebook interface includes a top bar with 'BDA_P9' and 'Ask Gemini' buttons, a file explorer, and a bottom status bar showing '28°C Sunny' and the date '04-10-2026'.

The image displays two sequential screenshots of a Jupyter Notebook interface, showing the process of generating a synthetic dataset.

Top Screenshot: The notebook is titled "BDA_P9" and shows the initial setup of data structures. The code defines three categories of hashtags, a list of platforms, and a list of regions.

```
hashtags = {  
    "Technology": [  
        "#AI",  
        "#MachineLearning",  
        "#Python",  
        "#CloudComputing",  
        "#DataScience"  
    ],  
    "Sports": [  
        "#Cricket",  
        "#Football",  
        "#IPL",  
        "#WorldCup",  
        "#Olympics"  
    ],  
    "Entertainment": [  
        "#Movies",  
        "#Music",  
        "#Bollywood",  
        "#Netflix",  
        "#Gaming"  
    ]  
}  
  
platforms = [  
    "Twitter",  
    "Instagram",  
    "Facebook",  
    "LinkedIn"  
]  
  
regions = [  
    "India",  
    "USA",  
    "UK",  
    "Canada",  
    "Australia"  
]
```

Bottom Screenshot: The notebook continues with the generation of 1000 synthetic posts. The code uses a loop to create a list of dictionaries, each representing a post with attributes like post_id, date, platform, hashtag, category, region, and engagement. The final output shows the first three rows of the resulting DataFrame.

```
data = []  
  
for i in range(1000):  
    category = random.choice(list(hashtags.keys()))  
    hashtag = random.choice(hashtags[category])  
  
    data.append({  
        "post_id": i + 1,  
        "date": fake.date_between(  
            start_date="-7d",  
            end_date="today"  
        ),  
        "platform": random.choice(platforms),  
        "hashtag": hashtag,  
        "category": category,  
        "region": random.choice(regions),  
        "engagement": random.randint(10, 10000)  
    })  
  
pdf = pd.DataFrame(data)  
pdf.head()
```

The output of the code is displayed as a table:

	post_id	date	platform	hashtag	category	region	engagement
0	1	2026-10-02	Facebook	#Bollywood	Entertainment	Australia	8459
1	2	2026-09-29	LinkedIn	#Music	Entertainment	India	9368
2	3	2026-10-02	Instagram	#Gaming	Entertainment	Canada	4547

The screenshot displays a Jupyter Notebook interface for a project named 'BDA_P9'. The notebook is running on a local host at `localhost:8888/notebooks/bda_p9%2fBDA_P9.ipynb`. The code in the notebook performs the following steps:

- Save a dataset to a CSV file named `social_media_hashtags.csv`.
- Load the dataset back into a Spark DataFrame using `spark.read.csv`.
- Display the first 10 rows of the dataset using `df.show(10, truncate=False)`.
- Print the schema of the DataFrame using `df.printSchema()`.
- Count the total records using `df.count()`.
- Calculate the sum of engagement for each row, handling null values, and display the results.

The output of the `df.show(10, truncate=False)` command is as follows:

post_id	date	platform	hashtag	category	region	engagement
1	2026-10-02	Facebook	#Bollywood	Entertainment	Australia	8459
2	2026-09-29	LinkedIn	#Music	Entertainment	India	9368
3	2026-10-03	Instagram	#Gaming	Entertainment	Canada	4547
4	2026-09-30	LinkedIn	#Olympics	Sports	UK	3224
5	2026-10-03	Twitter	#IPL	Sports	Canada	266

The output of the `df.printSchema()` command is as follows:

```
root
 |-- post_id: integer (nullable = true)
 |-- date: date (nullable = true)
 |-- platform: string (nullable = true)
 |-- hashtag: string (nullable = true)
 |-- category: string (nullable = true)
 |-- region: string (nullable = true)
 |-- engagement: integer (nullable = true)
```

The output of the `df.count()` command is as follows:

```
Total Records: 1000
```

The output of the `df.select(...).show()` command is as follows:

post_id	date	platform	hashtag	category	region	engagement
0						
1						
2						
3						
4						
5						
6						
7						
8						
9						

The image displays two screenshots of a Jupyter Notebook interface, likely running on a local host (localhost:8888). The notebook is titled "BDA_P9" and shows the results of data analysis on hashtags.

Top Screenshot: The notebook cell [10] displays the output of a query for hashtag frequency. The output is a table with columns 'hashtag' and 'frequency'.

hashtag	frequency
#Movies	85
#Cricket	75
#DataScience	73
#Bollywood	69
#Music	68
#AI	67
#WorldCup	66
#Gaming	66
#MachineLearning	65
#CloudComputing	64
#IPL	64
#Python	62
#Football	62
#Netflix	60
#Olympics	54

Bottom Screenshot: The notebook cell [12] displays the output of a query for hashtag engagement. The output is a table with columns 'hashtag', 'frequency', and 'average_engagement'.

hashtag	frequency	average_engagement
#Movies	85	5105.67
#Cricket	75	5280.95
#DataScience	73	4774.79
#Bollywood	69	4579.25
#Music	68	4611.49
#AI	67	5191.07
#WorldCup	66	4575.83
#Gaming	66	4288.06
#MachineLearning	65	5298.46
#CloudComputing	64	5531.86
#IPL	64	4841.34

The image displays two screenshots of a Jupyter Notebook interface, likely from a web browser. The browser address bar shows the URL `localhost:8888/notebooks/bda_p9%2fBDA_P9.ipynb`. The notebook is titled "BDA_P9" and shows the last checkpoint 14 minutes ago. The interface includes a menu bar (File, Edit, View, Run, Kernel, Settings, Help) and a toolbar with icons for file operations, running the code, and switching kernels. The code is written in Python and uses pandas for data manipulation.

Top Screenshot:

```
[13]: most_popular = hashtag_frequency.first()

print(
    "Most Popular Hashtag:",
    most_popular["hashtag"]
)

print(
    "Frequency:",
    most_popular["frequency"]
)
```

Most Popular Hashtag: #Movies
Frequency: 85

```
[14]: category_trends = df.groupby(
    "category"
).agg(
    count("total_posts"),
    round(avg("engagement"), 2).alias("average_engagement")
).orderBy(
    desc("total_posts")
)

category_trends.show()

+-----+-----+-----+
| category | total_posts | average_engagement |
+-----+-----+-----+
| Entertainment | 348 | 4672.93 |
| Technology | 331 | 5036.41 |
| Sports | 321 | 4987.22 |
```

Bottom Screenshot:

```
[15]: category_hashtags = df.groupby(
    "category",
    "hashtag"
).agg(
    count("frequency")
).orderBy(
    "category",
    desc("frequency")
)

category_hashtags.show(30, truncate=False)
```

category	hashtag	frequency
Entertainment	#Movies	85
Entertainment	#Bollywood	69
Entertainment	#Music	68
Entertainment	#Gaming	66
Entertainment	#Netflix	60
Sports	#Cricket	75
Sports	#WorldCup	66
Sports	#IPL	64
Sports	#Football	62
Sports	#Olympics	54
Technology	#DataScience	73
Technology	#AI	67
Technology	#MachineLearning	65
Technology	#CloudComputing	64
Technology	#Python	62

The image displays two screenshots of a Jupyter Notebook interface, showing the execution of Python code and the resulting data outputs.

Top Screenshot: The notebook shows the execution of code to analyze regional trends. The output displays a table of regional trends, sorted by frequency in descending order. The table shows the top 15 rows.

region	hashtag	frequency
Australia	#CloudComputing	21
Australia	#Cricket	18
Australia	#Movies	15
Australia	#WorldCup	15
Australia	#Bollywood	15
Australia	#Music	15
Australia	#DataScience	15
Australia	#Netflix	14
Australia	#Football	13
Australia	#Gaming	12
Australia	#MachineLearning	10
Australia	#Python	10
Australia	#IPL	10
Australia	#Olympics	8
Australia	#AI	8

Bottom Screenshot: The notebook shows the execution of code to analyze daily trends. The output displays a table of daily trends, sorted by frequency in descending order. The table shows the top 15 rows.

date	hashtag	frequency
2026-09-27	#Netflix	18
2026-09-27	#Cricket	17
2026-09-27	#Movies	16
2026-09-27	#Python	15
2026-09-27	#IPL	15
2026-09-27	#WorldCup	12
2026-09-27	#Olympics	10
2026-09-27	#DataScience	10
2026-09-27	#Bollywood	10
2026-09-27	#Gaming	9
2026-09-27	#CloudComputing	6
2026-09-27	#MachineLearning	6
2026-09-27	#AI	6
2026-09-27	#Music	6
2026-09-27	#Football	4

Below the table, the code identifies the top 3 trending hashtags:

```
[18]: top_3 = hashtag_frequency.limit(3)
print("Top 3 Trending Hashtags:")
top_3.show()
```

The output shows the top 3 trending hashtags:

hashtag	frequency
#Movies	85
#Cricket	75

Jupyter Notebook: BDA_P9

File Edit View Run Kernel Settings Help

Python 3 (ipykernel) Anaconda Toolbox

Cell [19]:

```
trend_score = df.groupby(
    "hashtag"
).agg(
    count("frequency").alias("frequency"),
    round(avg("engagement"), 2).alias("average_engagement")
).withColumn(
    "trend_score",
    round(
        col("frequency") * col("average_engagement"),
        2
    )
).orderBy(
    desc("trend_score")
)

trend_score.show()
```

hashtag	frequency	average_engagement	trend_score
#Movies	85	5105.67	433981.95
#Cricket	75	5280.95	396071.25
#CloudComputing	64	5531.86	354039.04
#DataScience	73	4774.79	348559.67
#AI	67	5191.07	347801.69
#MachineLearning	65	5298.46	344399.9
#Bollywood	69	4579.25	315968.25
#Music	68	4611.49	31581.32
#IPL	64	4841.34	309845.76
#WorldCup	66	4575.83	302004.78
#Olympics	54	5494.3	296692.2

Cell [20]:

```
top_trend = trend_score.first()

print("Overall Trending Hashtag:", top_trend["hashtag"])
print("Frequency:", top_trend["frequency"])
print("Average Engagement:", top_trend["average_engagement"])
print("Trend Score:", top_trend["trend_score"])
```

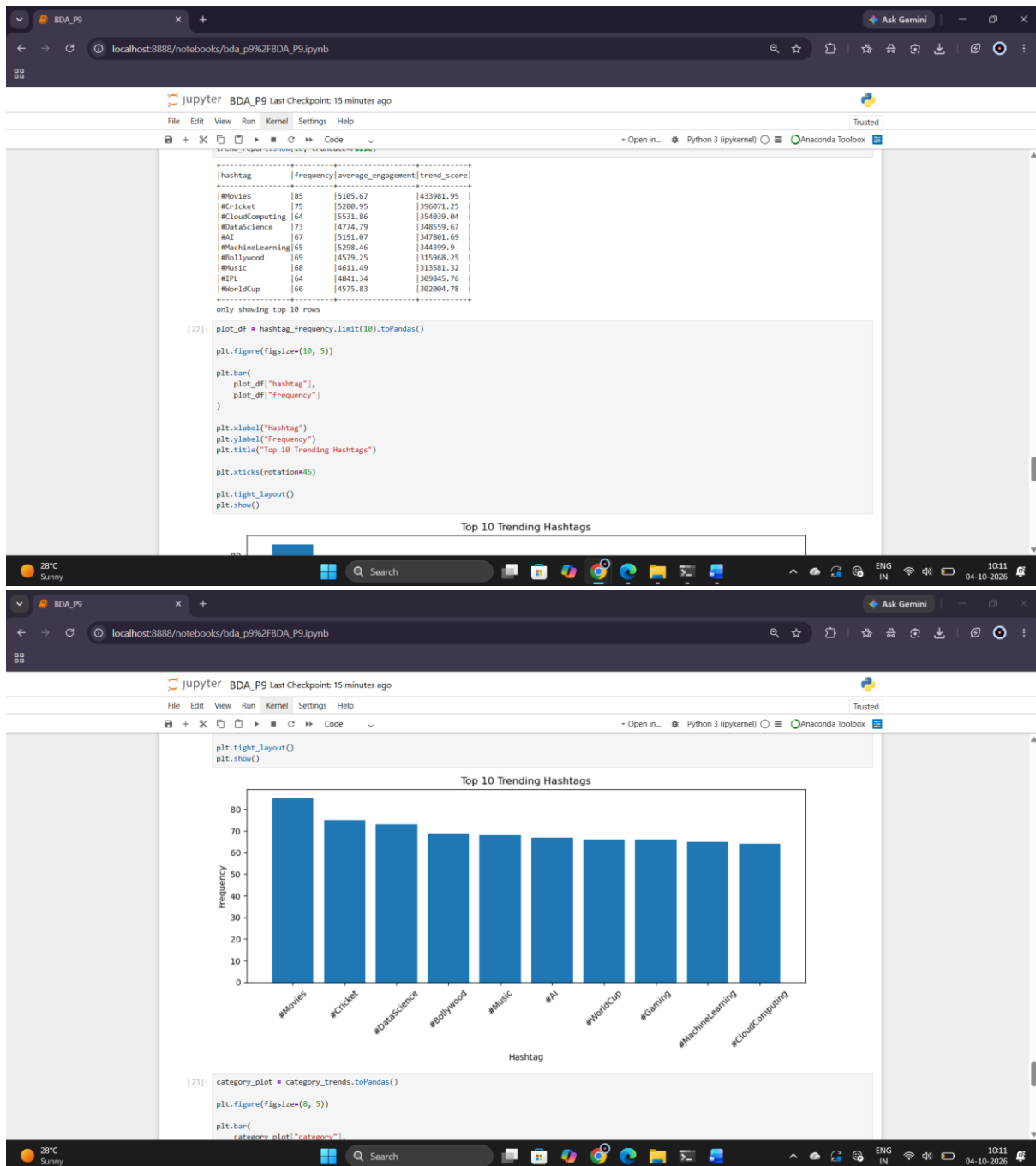
Overall Trending Hashtag: #Movies
Frequency: 85
Average Engagement: 5105.67
Trend Score: 433981.95

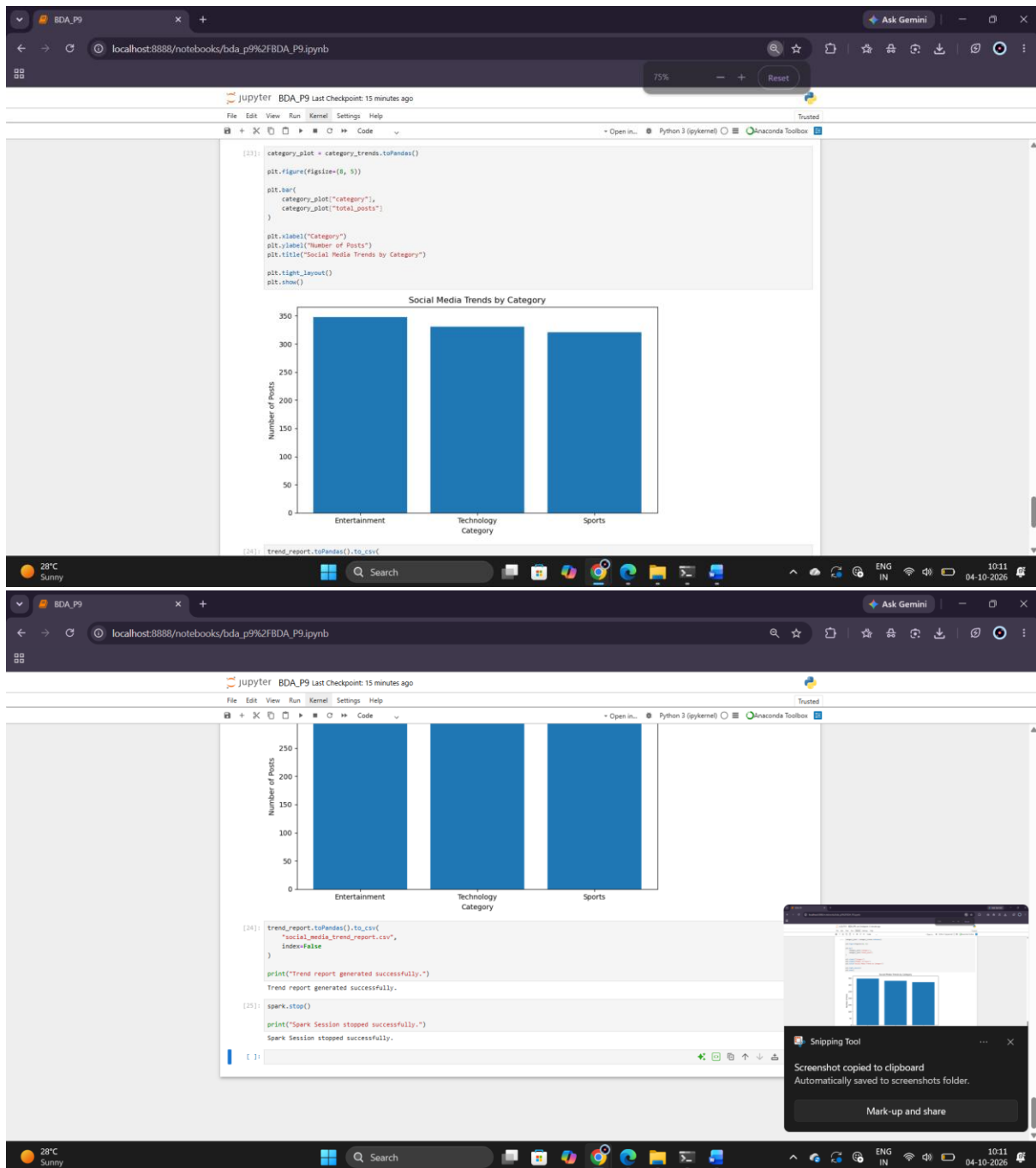
Cell [21]:

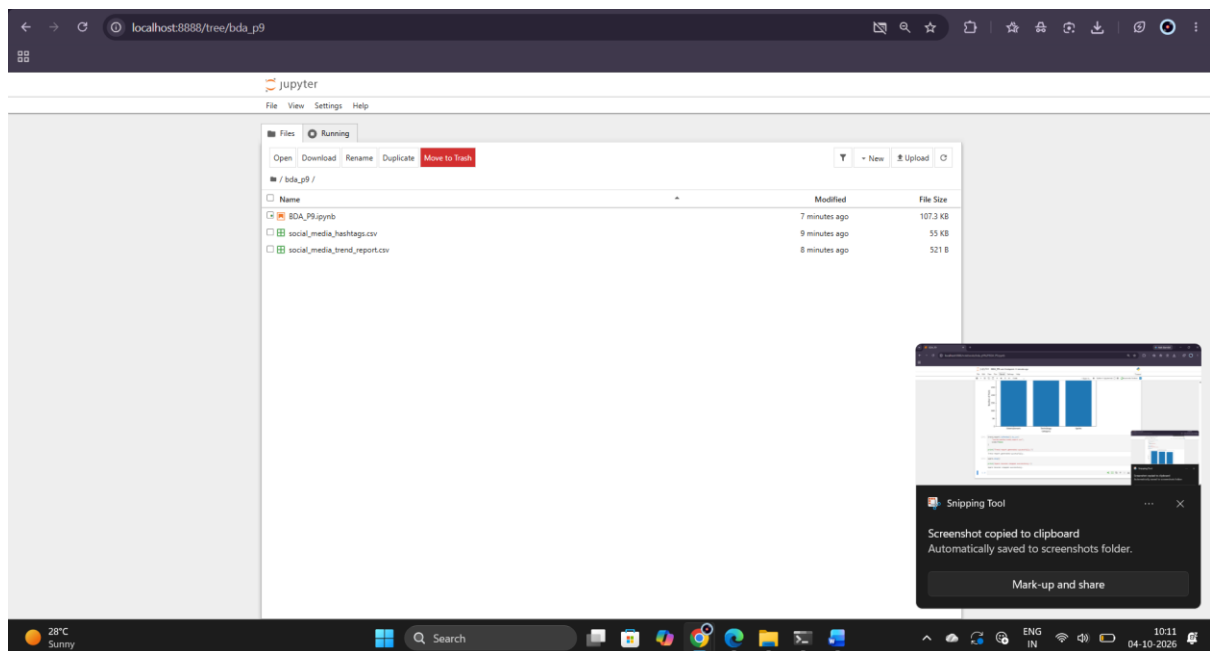
```
trend_report = trend_score.select(
    "hashtag",
    "frequency",
    "average_engagement",
    "trend_score"
)

trend_report.show(10, truncate=False)
```

hashtag	frequency	average_engagement	trend_score
#Movies	85	5105.67	433981.95
#Cricket	75	5280.95	396071.25
#CloudComputing	64	5531.86	354039.04
#DataScience	73	4774.79	348559.67
#AI	67	5191.07	347801.69
#MachineLearning	65	5298.46	344399.9
#Bollywood	69	4579.25	315968.25
#Music	68	4611.49	31581.32
#IPL	64	4841.34	309845.76







Key Questions / Analysis

Q1. Why is Apache Spark suitable for social media analytics?

Apache Spark is suitable for social media analytics because social media platforms generate very large volumes of data continuously.

Spark can process large datasets efficiently using distributed and parallel processing. Spark DataFrames provide convenient operations for filtering, grouping, aggregation, and analysis of social media records.

Spark can perform operations such as:

1. Hashtag frequency calculation
2. Engagement analysis
3. Category-wise analysis
4. Regional trend analysis

5. Daily trend analysis
6. Identification of popular hashtags
7. Trend detection

Spark can therefore help businesses analyze large-scale social media data efficiently and identify important trends.

Q2. What is the significance of hashtag frequency?

Hashtag frequency represents the number of times a particular hashtag appears in the social media dataset.

A higher hashtag frequency generally indicates that the topic is receiving greater attention or is being discussed by more users.

For example, if:

#AI 150 posts

#Cricket 120 posts

#Movies 95 posts

then #AI has the highest frequency in the dataset.

Hashtag frequency helps businesses:

1. Identify popular topics.
2. Detect trending subjects.
3. Understand user interests.
4. Compare different topics.
5. Plan relevant marketing campaigns.

In PySpark, hashtag frequency can be calculated using:

```
df.groupBy("hashtag").count().show()
```

Q3. How can trend analysis support digital marketing campaigns?

Trend analysis helps digital marketers understand what topics and hashtags are currently receiving attention from users.

Businesses can use trending information to create relevant content and marketing campaigns.

Trend analysis can help marketers:

1. Identify popular hashtags.
2. Understand customer interests.
3. Select suitable topics for campaigns.
4. Improve social media engagement.
5. Identify emerging trends.
6. Target specific regions or audiences.
7. Respond quickly to changes in user interests.

For example, if a technology-related hashtag becomes highly popular, a technology company can create social media content related to that topic.

Q4. How do businesses monitor social media trends?

Businesses monitor social media trends by collecting and analyzing data from different social media platforms.

They can analyze:

- Hashtags
- Posts
- Likes
- Comments
- Shares
- Views
- Engagement
- Mentions
- Categories
- Regions
- Trends over time

Apache Spark can process large amounts of social media data and perform aggregation operations to identify popular hashtags and emerging topics.

Businesses can then use these results for marketing, customer analysis, product planning, and decision making.

Q5. How can real-time analytics improve decision making?

Real-time analytics allows businesses to analyze incoming social media data as it is generated. Instead of waiting for a large batch of data to be processed, businesses can identify important changes quickly.

Real-time analytics can help businesses:

1. Detect emerging trends.
2. Monitor customer reactions.
3. Identify viral content.
4. Track hashtag popularity.
5. Detect sudden changes in engagement.
6. Respond quickly to customer feedback.
7. Improve marketing decisions.
8. Monitor campaigns in real time.

For real-time social media applications, Spark Structured Streaming can be used to process continuously arriving data and perform analytics as new records are received.

Supplementary Problems

1. Identify top 3 trending hashtags.
2. Compare trends across multiple days.
3. Analyze hashtag popularity by region.
4. Create hashtag categories (Technology, Sports, Entertainment).
5. Generate trend visualization using charts.

Conclusion

The Social Media Trend Detection Using PySpark practical was successfully completed using Apache Spark and PySpark.

A social media hashtag dataset was created/loaded into a Spark DataFrame, and its records and schema were examined. The dataset was validated and hashtag frequency was calculated using Spark DataFrame aggregation operations.

The hashtags were analyzed according to their frequency, category, region, and date. The top trending hashtags were identified, and trends across multiple days were compared. Technology, Sports, and Entertainment categories were also analyzed to understand the popularity of different types of social media content.

The practical demonstrated how Spark DataFrames, `groupBy()`, `count()`, aggregation functions, filtering, sorting, and distributed processing can be used for large-scale social media analytics and trend detection.

The analysis can help businesses identify popular topics, understand user interests, support digital marketing campaigns, and improve decision making using social media data.